Tribhuvan University

Faculty of Humanities and Social Science

A PROJECT REPORT ON

Athletix : A Football Team Management System

Submitted to

Department of Computer Application

Mechi Multiple Campus

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

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Mechi Multiple Campus

Supervisor's Recommendation

I hereby recommend that this project is prepared under my supervision by Bishal Pulami and Diwash Nepal entitled "Athletix : A Football Team Management System" in partial fulfillment of the requirements for the degree of Bachelor of computer Application is recommended for the final evaluation.

………………

SIGNATURE

Sunil Sharma

SUPERVISOR

Head of the Department

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LETTER OF APPROVAL

This is to certify that this project prepared by Bishal Pulami and Diwash Nepal entitled "Athletix : A Football Team Management System" in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

……………………………….. **Supervisor Sunil Sharma** **HOD (Computer Science and Application)** Bachelor's in Computer Application Mechi Multiple Campus	………………………………………. **Director Raju Poudel** Bachelor's in Computer Application Mechi Multiple Campus
…………………………………. **Internal Examiner**	………………………………. **External Examiner**

Acknowledgement

On the occasion of accomplishment of our project, we would like to express our sincere thanks to our Supervisor **Mr. Sunil Sharma** for his valuable guidance and support in completing our project.

We would like to express our gratitude towards our Program Director **Mr. Raju Poudel** for giving us this great opportunity to do a project on "**Athletix : A Football Team Management System**". Without their support and suggestions, this project would not have been completed. Their valuable guidance and suggestions helped us in various phases of the completion of this project. We will be thankful to you in this regard.

We are also thankful for seniors, other faculty and staff member of our department for kind cooperation and help. Many people have helped us during the completion of our report, and we would like to take this opportunity to show our gratitude and respect to all those who made this project possible for us.

Bishal pulami

Diwash Nepal

Abstract

Managing a football team today involves many tasks like planning matches and practices, tracking player attendance, and sending updates to players and staff. **Athletix: A Football Team Management App** is a simple mobile app made to help coaches, managers, and players handle all these activities in one place. It combines features like scheduling, attendance tracking, messaging, and performance tracking into a single platform, so there is no need to use many different tools or manual methods.

The app allows users to organize games and training sessions, keep track of who is present, send real-time messages, send notifications about the events and games, and view player performance using simple graphs with machine learning support. It is developed using Android Studio with Java and XML. Firebase is used for real-time data and login features, while SQLite is used to save data when there is no internet.

Athletix is built after studying popular apps like TeamSnap, Heja, and Sportlyzer. It focuses on solving real problems faced in managing football teams. Its clean design and flexible system make it useful for all kinds of teams — from small local clubs to professional groups.

This project highlights our skills in mobile app development and practical problem-solving. Athletix helps football teams save time, stay organized, and improve both communication and performance

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Keywords

HTML = HyperText Markup Language

CSS = Cascading StyleSheet

JS = Javascript

Firebase

FireStore Database

XGBoost

Isolation Forest

Chapter 1: Introduction

1.1. **Introduction**

Athletix is a complete solution for any football team, whether it's a local youth league, a recreational club, or a professional team. It brings together all the important tools you need in one place, so you don't have to juggle multiple apps or systems. This includes scheduling games and practices, sending reminders and updates to players, sharing football news, and keeping everyone connected through team communication tools.

Managing a football team involves numerous tasks, from scheduling games and practices to tracking player attendance and communicating with managers, coaches, players and other staffs. Our football team management app, Athletix is designed to simplify these responsibilities, empowering teams to operate more efficiently while fostering better communication and collaboration.

The Football Team Management System is all about making life easier for football teams by keeping things organized, clear, and accessible. It’s a step forward in modernizing how football teams are managed, helping teams perform better both on and off the field. Coaches and team organizers can spend more time supporting their players and helping them grow, instead of getting overwhelmed by paperwork and administrative tasks.

As technology continues to shape the future of sport football, our app stands as a testament to innovation and collaboration, helping teams around the world achieve their full potential. With its intuitive design and comprehensive features, it's the perfect solution for anyone looking to elevate their team's performance and make the football experience more enjoyable for everyone involved.

1.2.**Problem Statement**

The current methods of managing football teams often rely on manual processes or fragmented systems, which lead to inefficiencies and challenges for coaches, players, and administrators. These outdated approach create several issues that impact the smooth functioning of teams and leagues such as:

- **Unorganized Scheduling:** Without a centralized system, scheduling games, practices, and tournaments becomes chaotic. Important events can be missed or double-booked, causing confusion and frustration for everyone involved.

- **Poor Communication:** Coaches and team administrators often struggle to keep players, coaches, and staff informed about updates or changes. Disconnected communication channels like emails, phone calls, or group chats can lead to missed messages and misunderstandings.
- **Time-Consuming Administration:** Managing player registrations, players attendance and performance data manually takes a lot of time and effort. This leaves less time for coaches to focus on training and player development.
- **Lack of Transparency:** Without clear records of schedules or player performance, accountability is reduced. This can lead to disputes or dissatisfaction among team members.
- **Fragmented Data Management:** Player information, team rosters, and performance stats are often scattered across multiple platforms or paper files. This makes it difficult to access the right information quickly when needed. These challenges not only waste time but also hinder the overall experience for players, staffs, and coaches. A more streamlined and efficient solution is essential to ensure teams can focus on what truly matters—building skills, fostering teamwork, and enjoying the game.

1.3. Objectives

The main objectives of this project are listed below:

- To enable easy registration and tracking of player profiles, including their statistics and availability.
- To improve communication between coaches and players through in-app messaging or updates.
- To schedule matches, practices, and team events with notification features.

1.4. Scope and Limitation

1.1.1. Scope:

- The application will be developed as an Android mobile app.
- It will be built using the Java programming language.
- Team announcements and updates will be managed within the app.
- The app will support both single and multiple team management.
- It is designed to be used by managers, coaches, and players.

1.1.2. Limitations:

- The first version of the app will function offline or with limited online features.
- The app will only support football teams.
- Other sports will not be supported in this version.

1.5. **Development Methodology**

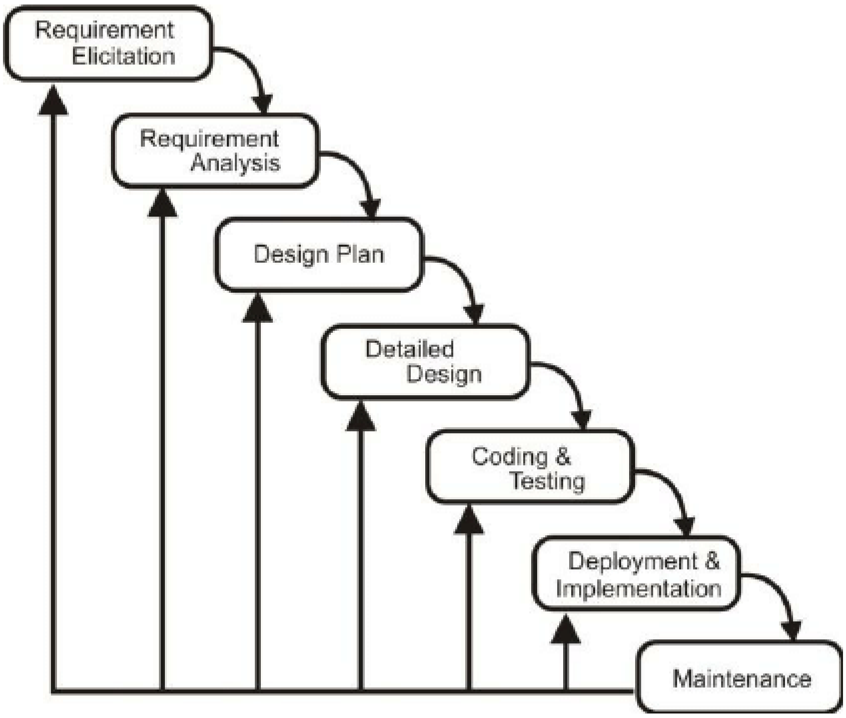


Figure 1: Iterative Waterfall Model

The **Iterative Waterfall Model** is a structured and disciplined Software Development Life Cycle (SDLC) approach that follows a sequential process while allowing feedback and improvements after each development cycle. Unlike the traditional waterfall model where each phase is completed once and for all, the iterative waterfall model introduces the concept of revisiting previous stages based on the insights gained in later phases. This makes it more flexible and adaptable than the classical waterfall methodology.

This model involves the following primary stages: **Requirement Analysis**, **System Design**, **Implementation**, **Testing**, **Deployment**, and **Maintenance**. However, in each iteration, the system is refined by revisiting earlier phases based on test results and feedback from stakeholders. Each cycle brings the product closer to its final, complete form, ensuring fewer errors, better quality, and more alignment with the user's expectations.

In the case of Athletix, this model was chosen because of its ability to combine structured planning with controlled iteration. The initial phase focused on gathering requirements from the perspective of football coaches, players, and managers. Based on this, a detailed system design was prepared including UI layout, data models, and component architecture.

The logical reason for choosing the Iterative Waterfall Model for Athletix lies in the project's moderately changing requirements and the need for systematic documentation. Since the overall scope was defined early on, and the majority of features were planned in advance, a structured approach ensured better time management, proper documentation, and clarity in the deliverables. At the same time, the ability to revise previous stages allowed improvements without starting from scratch.

This model also supports formal review sessions after each stage, ensuring that design flaws and implementation errors are caught early. As a result, the development process stayed controlled, traceable, and well-documented — essential for academic and professional standards.

This methodology proved to be highly effective for a mobile application like Athletix, where features were mostly pre-planned but still required refinement after practical testing and user feedback.

1.6. **Report Organization**

- **Chapter 1 – Introduction**

  This chapter introduces the project, with the problem statement, objectives, scope, and limitations. It clearly identifies the problems to be addressed by the project, the particular objectives to be achieved, and the range and boundaries of the area in which the project operates.

- **Chapter 2 – Background Study**

  This chapter gives a background study relevant to the project, along with a general overview of its parts and workings. Apart from this, it also contains a literature review for arriving at a broader conceptual understanding regarding concepts of the project. Further, it talks about other related systems already developed to compare with the proposed current project.

- **Chapter 3 – System Analysis and Design**

  This chapter focuses on the system analysis and design process via various charts and diagrams. It defines the system's functional specifications via use cases and other efficient techniques. Furthermore, it offers the database schema, interface design, and deployment diagram to clearly and systematically express the system.

- **Chapter 4 – Tools, Techniques, and Testing**

  This chapter introduces the techniques and tools used in carrying out the project. It also discusses how test cases used in testing the system, both at the unit level and as a whole system, were developed to ensure its functionality, reliability, and performance.

- **Chapter 5 – Lessons Learned, Recommendations, and Conclusion**

  This chapter has references to the lessons learned over the course of the project, from start to completion. It offers recommendations to subsequent projects of a similar nature as a means of doing things differently or avoiding potential problems. It then finally presents the overall conclusion, an encapsulation of the findings and significance of the project.

Chapter 2: Background Study and Literature Review

2.1. Background Study

Football is the world's most popular sport, with over 250 million active players and millions of teams at all levels—from grassroots to elite professional clubs. Managing these teams efficiently has become increasingly complex as team sizes grow, match frequency increases, and data becomes critical for coaching and decision-making.

Team management systems have emerged globally to simplify this complexity. These apps aim to centralize operations like scheduling, player registration, communication, and performance tracking. Some of the leading tools include **TeamSnap**, **SportEasy**, **Heja**, **Spond**, and **TeamLinkt**. The global market is moving toward digital transformation, with clubs adopting software to handle logistics, communication, match analytics, and financial operations. Even amateur clubs now face the pressure to adopt tech for improved coordination and professionalism.

Existing platforms cover basic needs like event scheduling, availability tracking, group messaging, and fee collection. However, many of them fall short when it comes to deeper functionality, integration, and affordability, especially for smaller or developing clubs. Some of the major features they are lacking are:

- Advanced performance analytics are largely missing
- Almost everything is done manually
- Offline support is poor in many apps
- Match preparation features are often underdeveloped
- Pricing is a significant barrier
- User interface and experience (UI/UX) issues are common
- Clubs currently use fragmented tools
- Emerging technologies are transforming the elite football landscape
- IoT and blockchain are entering the football tech space
- Grassroots teams remain underserved

Athletix: A Football Team Management System aims to close this gap by building a flexible, mobile-first app that is affordable, customizable, and includes both online and offline features. Athletix will combine the core features of established tools— scheduling, player profiles, attendance, and communication—with performance analytics, match planning, and formation design, supported by Firebase (real-time sync). It is designed using Flutter, this app will allow role-based access (coach, player, admin) and support integration with modern tools and analytics as the system evolves. By focusing on football specifically, Athletix addresses pain points ignored by general sports apps and high-end enterprise tools, offering a balanced, cost-effective platform for both grassroots and semi-professional teams.

2.2. Literature Review

Several football team management systems have been developed to address the growing administrative and performance-tracking needs of coaches, players, and clubs. These systems aim to streamline operations like scheduling, communication, and performance tracking. The following review highlights the core features, strengths, and limitations of some widely used systems in the global football ecosystem.

TeamSnap

TeamSnap is a general-purpose sports team management app offering scheduling, roster management, availability tracking, messaging, payment processing, and statistics in one platformteamsnap.com. It is widely used by coaches, parents, and administrators of youth and amateur teams (over 24 million people across 3 million teams, clubs, and leagues use itteamsnap.com). TeamSnap aims to simplify communication (e.g. team chat and notifications) and logistics (e.g. automated calendar syncing) for small clubs and recreational leagues. While its ease-of-use is often praised, some users report limitations: for instance, location maps can be inaccurate and billing/refund policies can be confusing. TeamSnap is not focused on professional-level analytics or player development features, so larger organizations may need more specialized systems. [1]

SportEasy

SportEasy is a sports team management app explicitly designed for *amateur* clubs and team. It provides a comprehensive toolset including member/roster management, group messaging, event calendars with attendance tracking, lineup planning, task assignments, and basic stats reporting. The platform markets itself as “the most complete app” for organizing teams in any sport (soccer, rugby, basketball, etc). SportEasy is aimed at club administrators, coaches and players in recreational or youth sports; its Terms of Use explicitly note it is “mainly dedicated to amateur sports and communities”. Because of this amateur focus, it lacks the enterprise features needed by professional organizations, and there are no reports of advanced analytics beyond basic game stats. [2]

Heja

Heja is a free team communication and scheduling app for youth sports. It emphasizes simplicity: coaches and parents can quickly set up games/practices, send RSVP availability requests, and use group chat for messaging (including photo/video sharing). Heja advertises itself as “the best app for soccer coaches” and has built a user base of over 500,000 teams. Its features include automated reminders and attendance tracking to reduce manual follow-up. Target users are typically youth or school teams where parents and volunteer coaches need a shared platform. Heja’s main limitation is that it is focused on basic organization and communication; it does not include advanced training or performance analysis tools. [3]

360Player

360Player (formerly Team Analytics) is an all-in-one club management platform aimed at *sports organizations*, including professional clubs. It handles member registrations, payments, club communications, scheduling, and offers player development modules. For example, the “Pro” version includes training session planning, video analysis, performance statistics, and player development tracking. In effect, 360Player connects club administration (registrations, websites, payments) with coach/athlete tools (training plans, data dashboards) within one “digital hub”. The platform is intended for medium to large clubs and is used by professional teams; one marketing page explicitly targets “professional clubs” seeking comprehensive player development and management solutions. Because of its depth, 360Player may be too complex or costly for casual teams. Smaller organizations might find its full feature set unnecessary, and no free tier is offered, so 360Player is best suited to well-resourced clubs. [4]

Spond

Spond is a mobile app designed to simplify team/group management for community sports and clubs. It is essentially an all-in-one solution: organizers can create groups (up to 500 members), send event invites and automated reminders, collect payments, store team files securely, and use a central chat for communication. Spond is popular in youth and amateur sports across Europe. The basic Spond app is free and targeted at coaches, team managers, and parents who need to coordinate practices, games, or social events. (Larger organizations can upgrade to “Spond Club” for more advanced club-level tools.) Because its core is free group management, a limitation is that very large clubs or leagues may require the paid version to handle hundreds of players. Compared to enterprise software, Spond also has fewer specialized analytics or medical tracking features. [5]

TeamLinkt

TeamLinkt offers a free all-in-one platform aimed at leagues, clubs, and associations, with a particular focus on soccer. It provides online registration, club/league websites, and scheduling tools. Each team gains access to the TeamLinkt mobile app, which supports roster sharing, full season schedules, player availability/RSVPs, automated reminders, team chat, photo sharing, live video streaming, and even basic stats trackingteamlinkt.com. In short, TeamLinkt covers both administrative needs (registration, websites) and team- level communication/coordination in one system. The target users are youth and amateur clubs and leagues that want to manage a season for free. Since TeamLinkt’ s model is free, it may lack some advanced features found in paid platforms (e.g. deep analytics or dedicated support). Professional teams would likely outgrow its capabilities, as it does not include high-end performance modules. [6]

Kitman Labs

Kitman Labs provides a high-end athlete performance management platform rather than a basic team organizer. Its system consolidates player health and performance data (training load, game stats, medical/injury records, wellness surveys, etc.) into a unified analytics tool. It is used by professional and elite clubs (e.g. premier soccer and rugby clubs and leagues) to optimize player development and prevent injuries. Key features include injury lifecycle tracking, medical history management, training and session planning, and rich dashboards for performance analytics. Kitman is not a general team communication app – it does not handle scheduling games or messaging parents. Instead, it focuses on sports science and player profiling. As such, its limitations are that it is expensive and complex, intended only for well-resourced professional organizations. It has negligible appeal for youth or amateur teams. [7]

LaLiga Mediacoach

Mediacoach is the proprietary video and data analysis platform used by Spain's top professional league (LaLiga). Developed in-house, it serves all 42 LaLiga clubs. Mediacoach ingests match video from an array of cameras (16 optical tracking cameras plus tactical video) and extracts huge amounts of data: on the order of 3.5 million data points per gam0065. It applies advanced algorithms and machine learning (e.g. via Microsoft Azure) to produce tactical, technical and physical metrics. Clubs and analysts use Mediacoach for opposition scouting, player workload monitoring, and performance optimization. Because it is a league-level system, Mediacoach democratizes access to advanced analytics across the league . However, it is exclusively for professional teams: it requires specialized hardware, expertise, and a license from LaLiga. Its limitation is obvious: no amateur club can access Mediacoach, and it has no functionality for basic team management (scheduling, parent communication, etc.). [8]

Chapter 3 : System Analysis and Design

3.1. System Analysis (Object-Oriented Approach)

The system analysis phase involves understanding the functional and non-functional expectations of the users and evaluating whether the system is technically and economically feasible to implement. As Athletix is developed using an Object-Oriented Approach, this chapter includes Object Modeling, Dynamic Modeling, and Process Modeling to describe system functionality and design.

3.1.1. Requirement Analysis

I. Functional Requirements:

Functional requirements describe what the system should do. The Athletix app provides the following major functionalities:

- User registration and login with authentication.
- Role-based access: Admin, Coach, Player.
- Create and manage football teams.
- Add and manage players under teams.
- Schedule events, matches and practice sessions.
- Track player attendance.
- View performance stats and visualize through charts.
- Real-time team chat.
- Send real-time push notifications.
- Store match history and player data offline using SQLite.

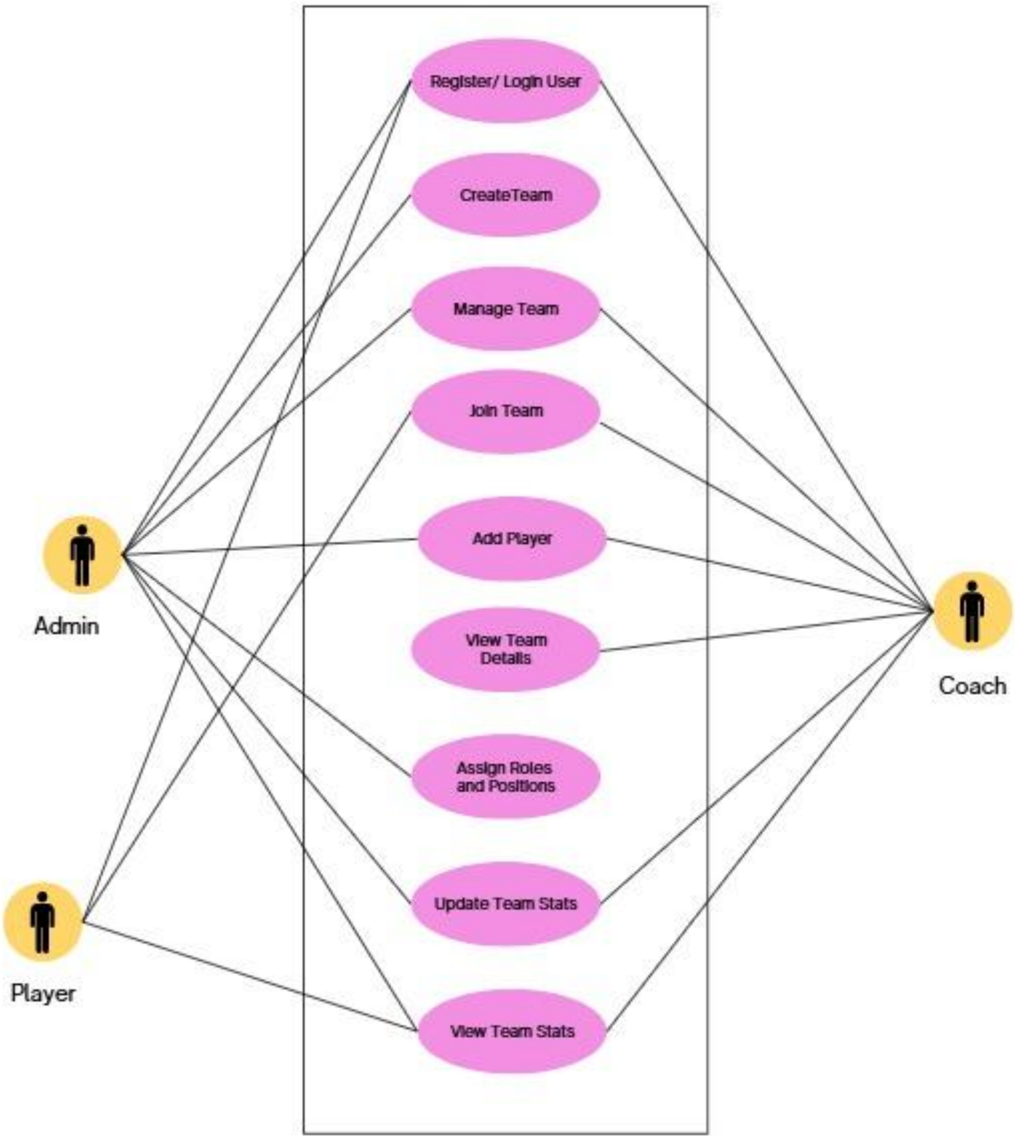


Figure 2: Use-Case Diagram of Athletix: A football Team Management System

Use-Case	**Description**
Register/Login	Users can register or log in using Authentication.
Create Team	Admins can create a new football team and assign players.
Manage team	Admins/Coaches can add and manage teams.
Join Team	Users can join team using team code.
View Performance	Players and coaches view match stats and training performance.
Add Player	Admin can add players.
Assign Roles	Admin can Assign Roles and Position.
Update Team Stats	Coach can update team stats.
View Team Stats	All users of the respective team can view their team stats.

ii. Non-Functional Requirements

- **Performance**: Web-app should operate smoothly with minimal loading time.
- **Scalability**: Firebase backend can handle multiple teams and users.
- **Usability**: Simple UI using HTML page suitable for coaches and players.
- **Security**: Firebase Authentication and secure data access.
- **Reliability**: Cloud (Firebase) must be consistent.

3.1.2 Feasibility Analysis

A feasibility study is crucial for our project as it evaluates the technical, operational, and economic viability of implementing Athletix. It helps us assess the availability of resources, integration into existing processes, and financial feasibility. By conducting this study, we can decide if the project can move forward or not.

a. Technical Feasibility

Tools Used: HTML, JS, Firebase

- Firebase provides real-time data sync, authentication, and cloud messaging.
- HTML, JS tools is used to develop frontend and backend of the system.
- All technologies are open-source or support free tiers for students.

b. Operational Feasibility

- Easy for coaches and players to use.
- Reduces manual scheduling and communication efforts.
- Saves time and ensures fewer missed matches or training.

c. Economic Feasibility

- No hardware investment required.
- Firebase offers sufficient free-tier limits.
- Development using free tools keeps the cost within student budget.
- Long-term savings by reducing manual administration and errors.

3.1.3. Object Modeling : Object and Class Diagram

Class Diagram

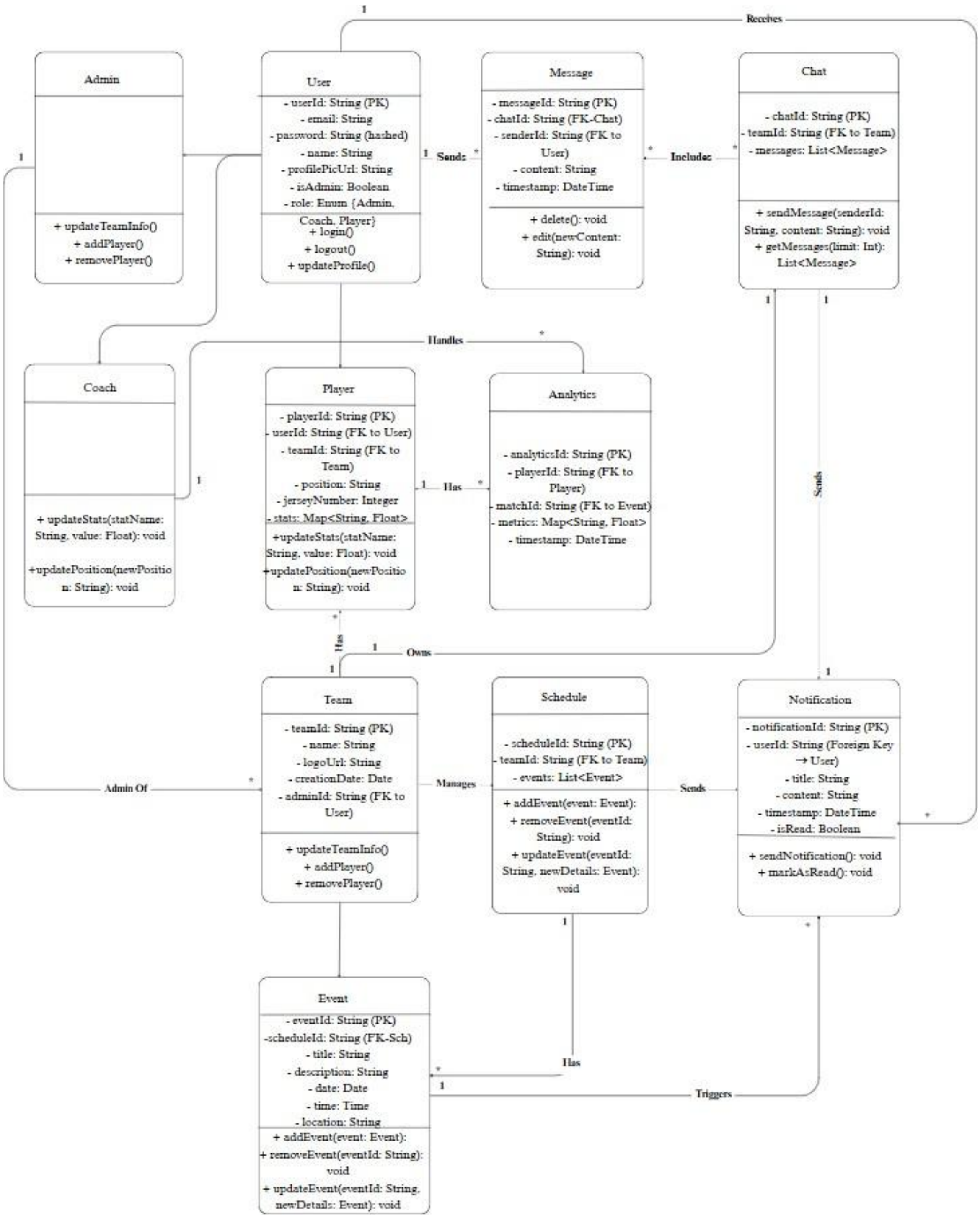


Figure 3: UML Class Diagram

3.1.4 Dynamic Modeling

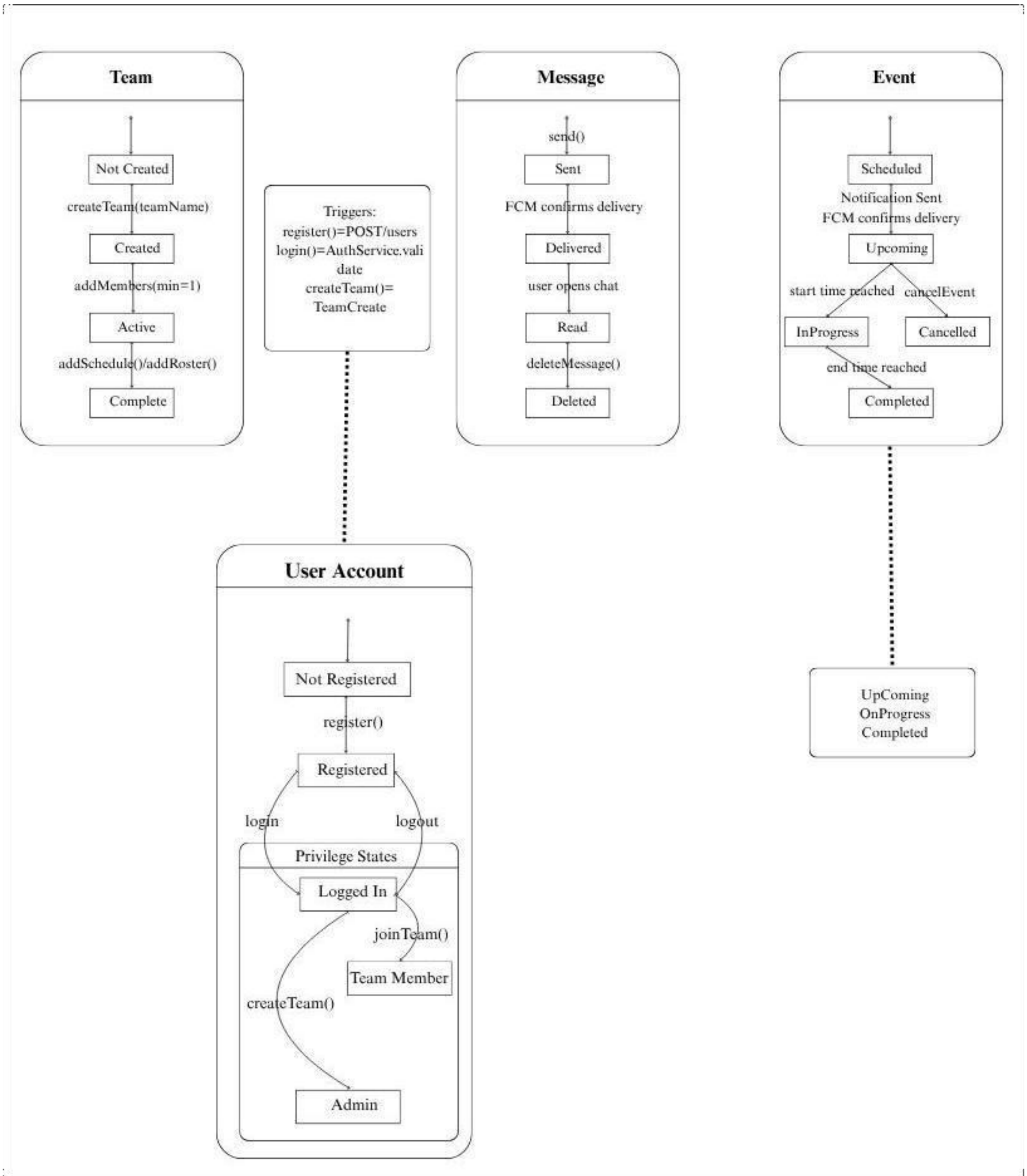


Figure 4: State Diagram

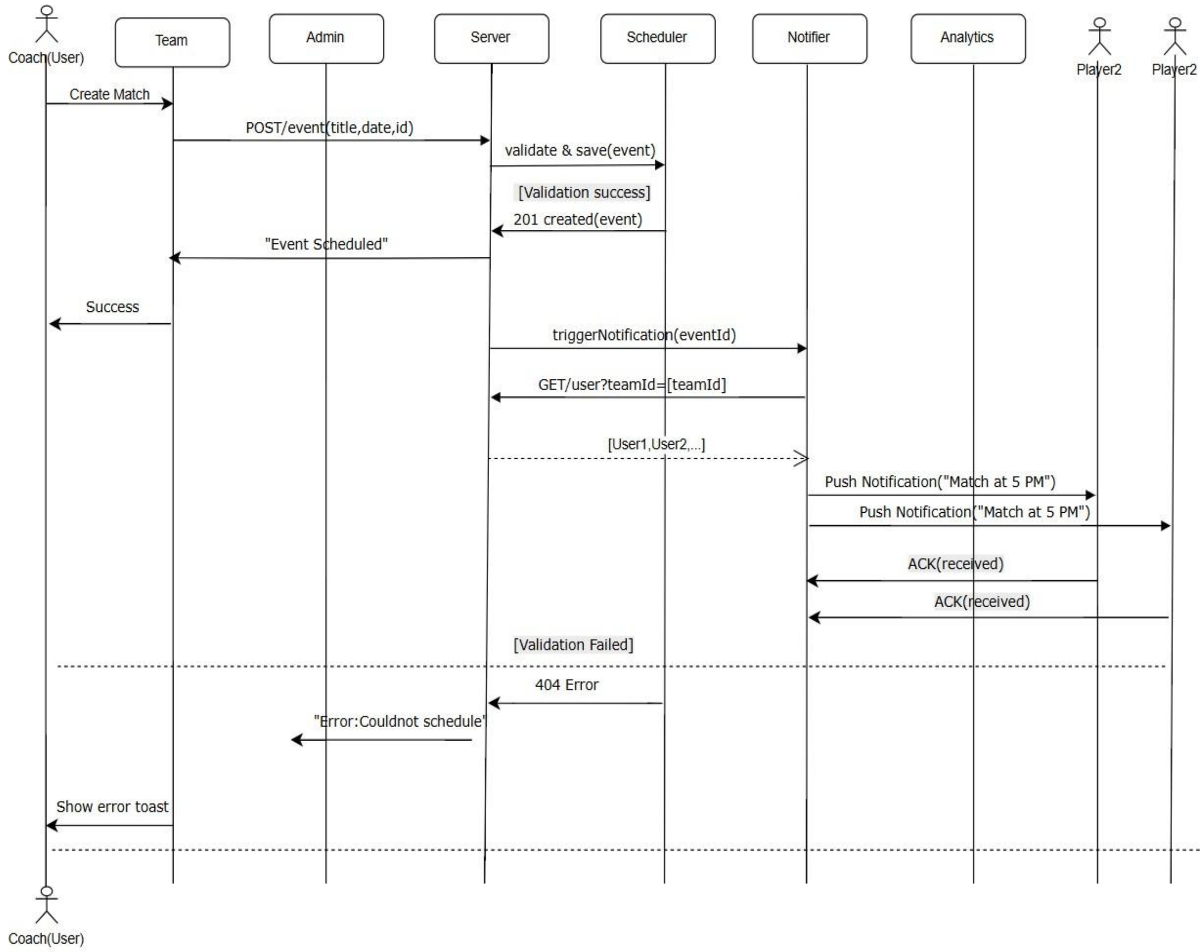


Figure 5: Sequence Diagram

3.1.4 Process Modelling:

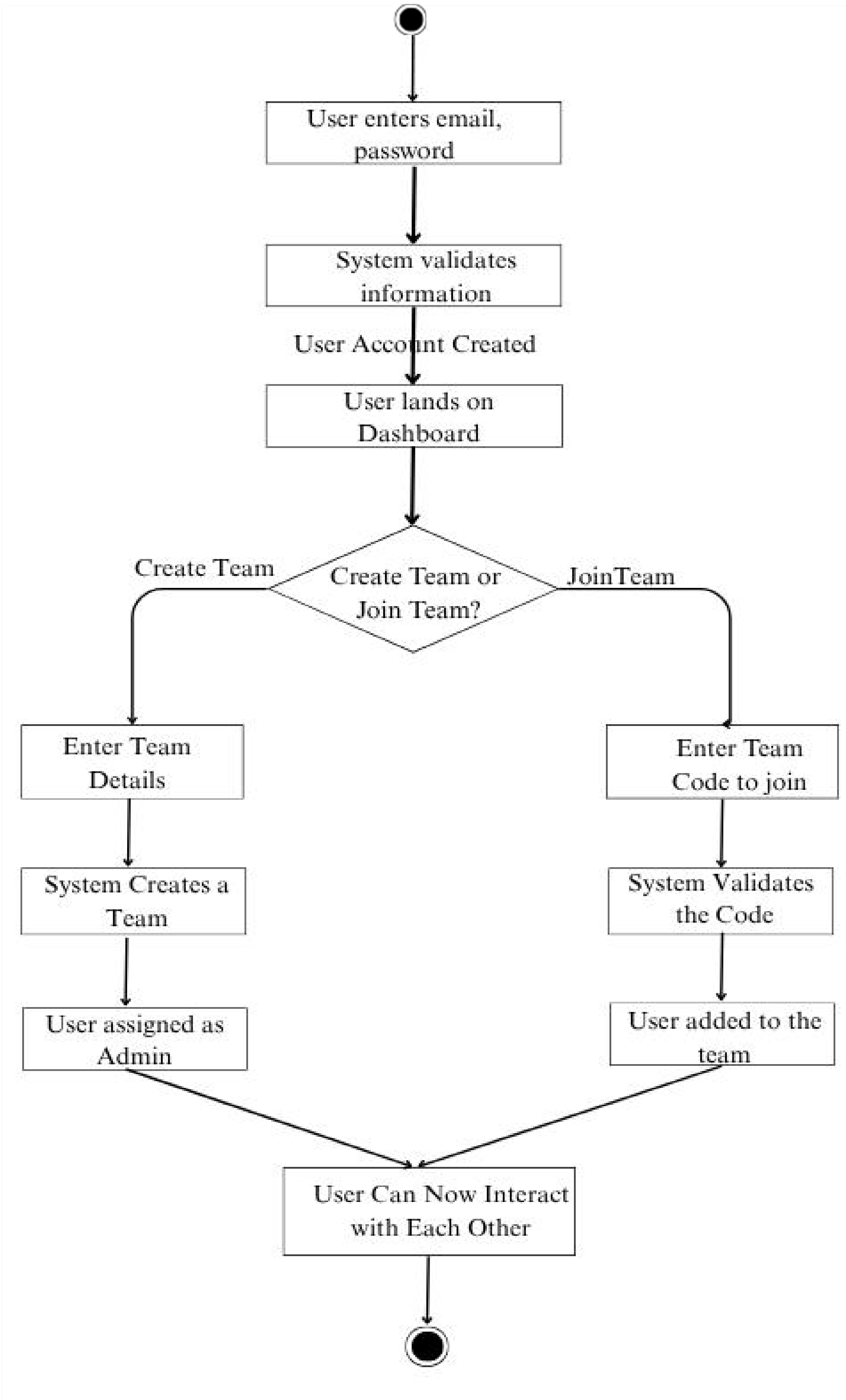
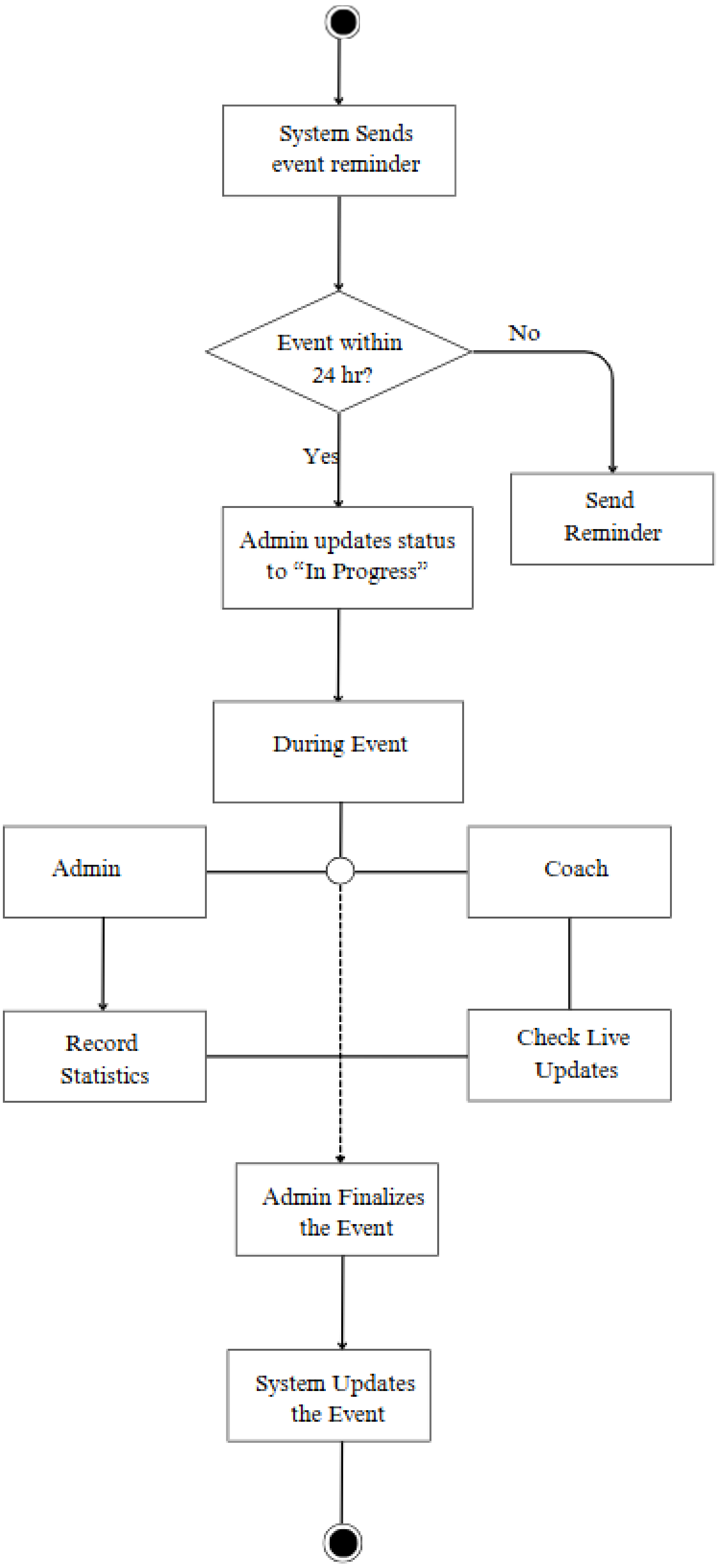


Figure 6: Activity Diagram


System Sends event reminder
Event within 24 hr?
No
Yes
Send Reminder
Admin updates status to “In Progress”
During Event
Admin
Coach
Record Statistics
Check Live Updates
Admin Finalizes the Event
System Updates the Event

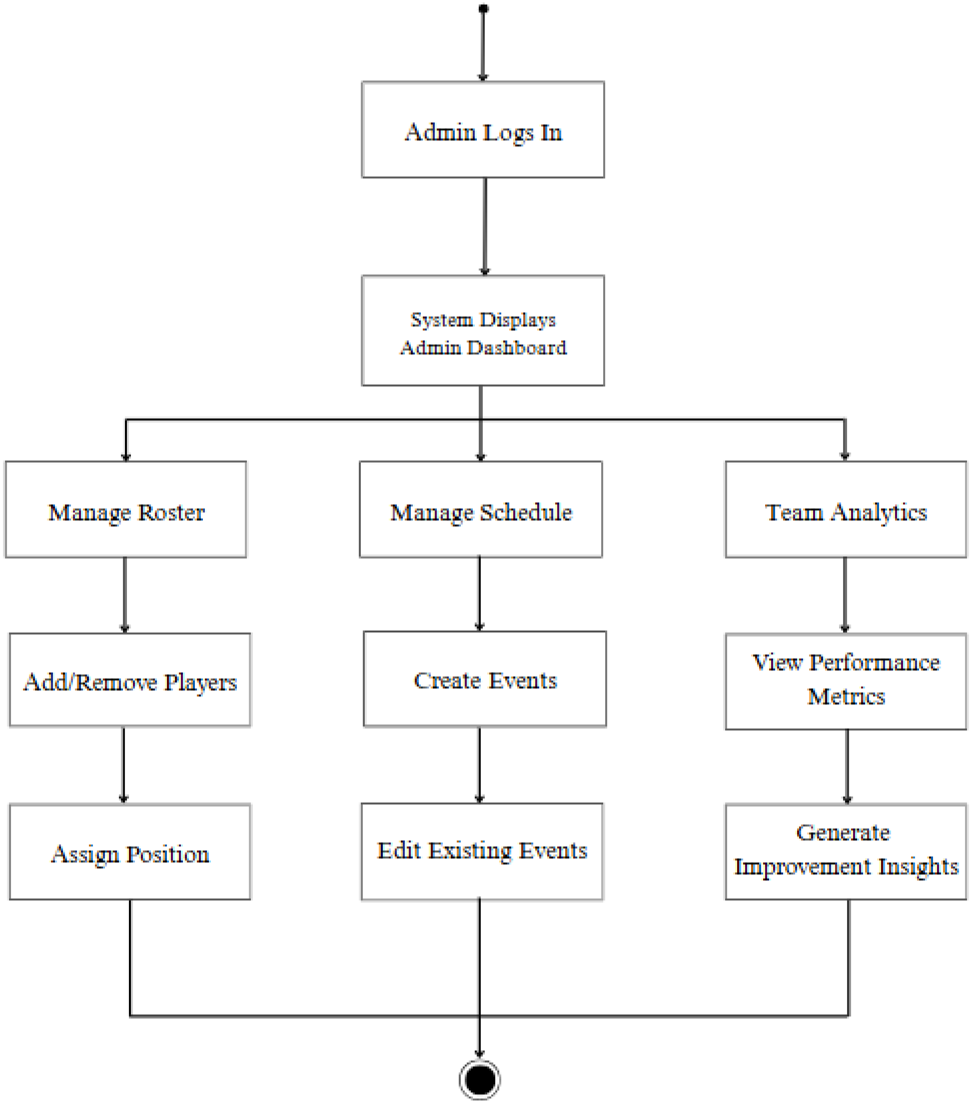


Figure 7: GameDay Workflow

3.2 System Design (Object-Oriented Approach)

3.2.1 Refinement of Classes

Each object defined earlier is refined to include detailed attributes and methods:

- **Coach**: Can add team members, manage players, schedule matches.
- **Player**: Can view schedules, mark attendance, view stats.
- **Match**: Stores data like date, time, opponent, venue.
- **Performance**: Records goals, assists, cards, fitness rating.

- **Team**: Links all players and coach; manages related data.

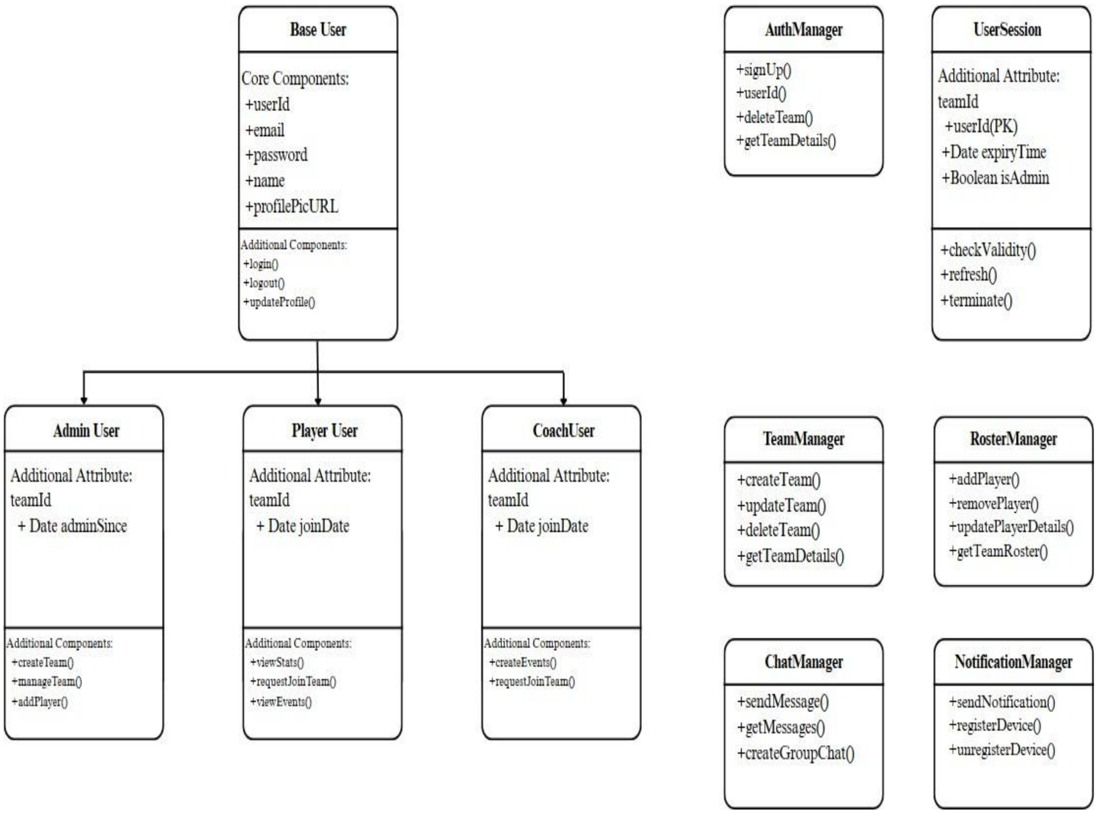


Figure 8: System Design

3.2.2 Component Diagram

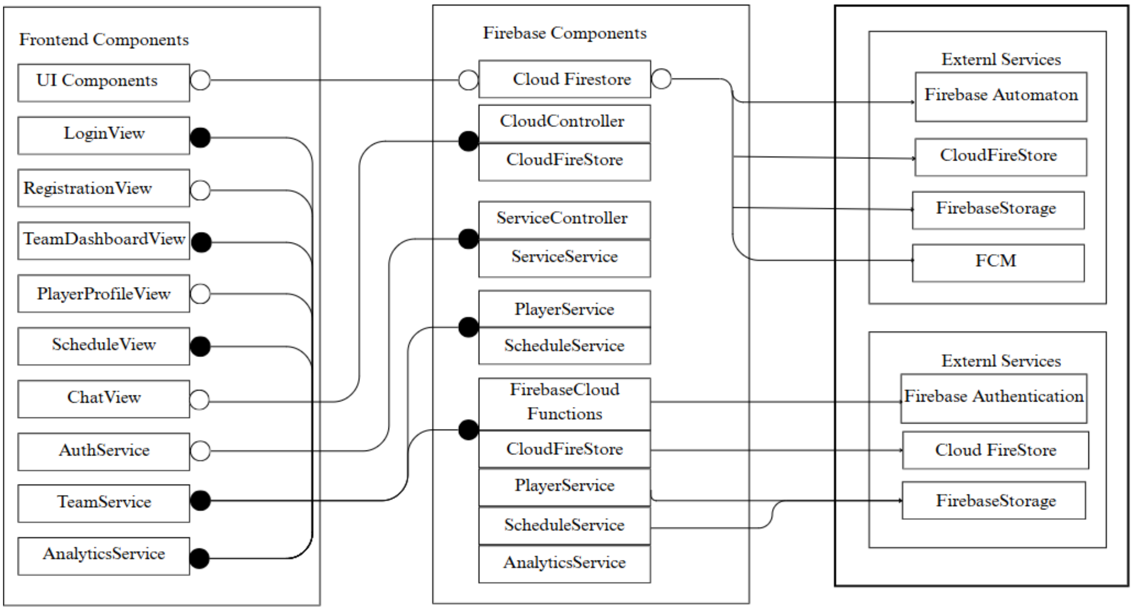


Figure 9: Component Diagram

3.2.3 Deployment Diagram

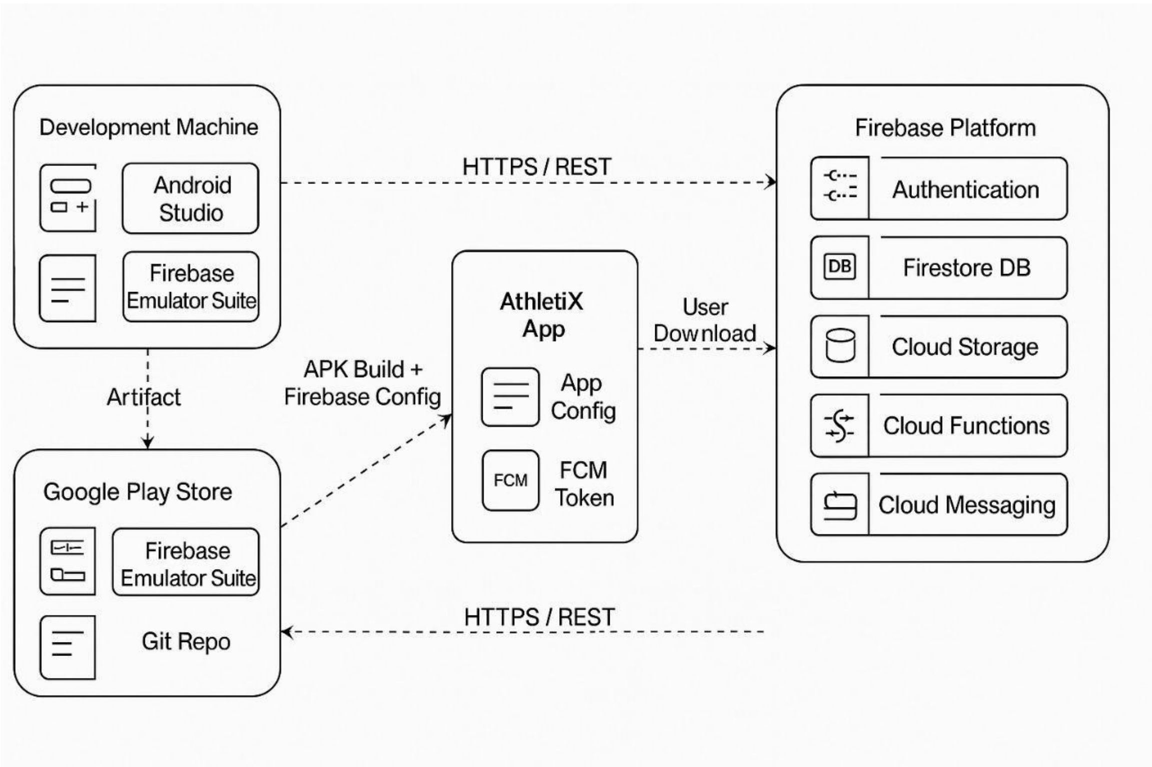


Figure 10: Deployment Diagram

3.3. Algorithm Details:

Gradient Boosting Regressor (GBR) / XGBoost

Gradient Boosting Regressor (GBR), including its optimized variant XGBoost, is an ensemble machine learning algorithm that builds decision trees sequentially, where each new tree corrects the errors of the previous one. Unlike Random Forest (which trains trees independently), GBR iteratively minimizes prediction errors using gradient descent, making it highly accurate for structured datasets.

The following steps show how it works:

- **Initial Prediction**: Starts with a simple model (e.g., mean player rating).
- **Error Calculation**: Measures residuals (actual vs. predicted).
- **Sequential Tree Building**: Each new tree predicts the residuals, refining accuracy.
- **Final Prediction**: Sum of all tree predictions.

It is ideal for football player performance analysis due to following reasons:

- It handles complex relationships.
- It captures non-linear patterns (e.g., a midfielder with low goals but high pass accuracy may still be rated highly).
- It recognizes interactions between features (e.g., "tackles + interceptions" matter more for defenders).
- It is optimized for medium-sized datasets

```
import numpy as np,pandas as pd,xgboost as xgb,joblib,os
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
class XGBoostPredictor:
    def __init__(self):
        self.model=None
        self.position_encoder=LabelEncoder()

self.feature_columns=['minutes_played','goals','assists','passes_completed','passes_attempted','tackles_won','tackles_attempted','shots_on_target','shots_total','interceptions','clearances']
        self.position_encoded=False
```

```
        self.load_or_create_model()
    def load_or_create_model(self):
        m_path='xgboost_model.pkl';e_path='position_encoder.pkl'
        if os.path.exists(m_path) and os.path.exists(e_path):
            try:
                self.model=joblib.load(m_path)
                self.position_encoder=joblib.load(e_path)
                self.position_encoded=True;print("Loaded existing XGBoost model");return
            except:pass
        self.create_training_data_and_train()
    def create_training_data_and_train(self):
        print("Creating synthetic training data...")
        np.random.seed(42);n=1000

data={'minutes_played':np.random.randint(10,90,n),'goals':np.random.poisson(0.3,n),'assists':np.random.poisson(0.2,n),'passes_completed':np.random.randint(10,100,n),'passes_attempted':np.random.randint(15,120,n),'tackles_won':np.random.randint(0,10,n),'tackles_attempted':np.random.randint(0,15,n),'shots_on_target':np.random.randint(0,8,n),'shots_total':np.random.randint(0,12,n),'interceptions':np.random.randint(0,8,n),'clearances':np.random.randint(0,15,n),'position':np.random.choice(['Forward','Midfielder','Defender','Goalkeeper'],n)}
        df=pd.DataFrame(data)
        df['passes_attempted']=np.maximum(df['passes_attempted'],df['passes_completed'])
        df['rating']=self.calculate_synthetic_rating(df)
        X=df[self.feature_columns+['position']].copy()
        self.position_encoder.fit(['Forward','Midfielder','Defender','Goalkeeper'])

X['position_encoded']=self.position_encoder.transform(X['position']);X=X.drop('position',axis=1)
        self.position_encoded=True;y=df['rating']
        X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)

self.model=xgb.XGBRegressor(n_estimators=100,max_depth=6,learning_rate=0.1,random_state=42)
        self.model.fit(X_train,y_train)
```

```
joblib.dump(self.model,'xgboost_model.pkl');joblib.dump(self.position_encoder,'position_encoder.pkl')
        print(f"Model trained. Train R²:{self.model.score(X_train,y_train):.3f}, Test R²:{self.model.score(X_test,y_test):.3f}")
    def calculate_synthetic_rating(self,df):
        r=np.zeros(len(df))
        for i,row in df.iterrows():
            base=min(row['minutes_played']/90*30,30)
            g=row['goals']*15;a=row['assists']*10
            pass_rating=(row['passes_completed']/row['passes_attempted']*20) if row['passes_attempted']>0 else 0
            def_rating=(row['tackles_won']/row['tackles_attempted']*15+row['interceptions']*2) if row['tackles_attempted']>0 else row['interceptions']*2
            shot_rating=(row['shots_on_target']/row['shots_total']*10) if row['shots_total']>0 else 0
            if row['position']=='Forward':r[i]=base+g+a+pass_rating*0.5+shot_rating
            elif row['position']=='Midfielder':r[i]=base+g*0.7+a+pass_rating+def_rating*0.5
            elif row['position']=='Defender':r[i]=base+g*0.3+pass_rating*0.7+def_rating+row['clearances']*1.5
            else:r[i]=base+pass_rating*0.3+def_rating
            r[i]+=np.random.normal(0,5);r[i]=max(0,min(100,r[i]))
        return r
    def predict(self,features):
        if self.model is None:raise ValueError("Model not trained")
        v=[features.get(c,0) for c in self.feature_columns]
        v.append(self.position_encoder.transform([features['position']])[0] if 'position' in features and self.position_encoded else 0)
        return max(0,min(100,float(self.model.predict(np.array(v).reshape(1,-1))[0])))
    def retrain_with_new_data(self,records):
        if not records:return

data=[{'minutes_played':r.minutes_played,'goals':r.goals,'assists':r.assists,'passes_completed':r.passes_completed,'passes_attempted':r.passes_attempted,'tackles_won':r.tackles_won,'tackles_attempted':r.tackles_attempted,'shots_on_target':r.shots_on_target,'shots_total':r.shots_total,'interce
```

```
ptions':r.interceptions,'clearances':r.clearances,'position':r.player.position,'rating':r.predicted_rating} for r in records if r.predicted_rating is not None]
        if len(data)<10:print("Not enough data");return
        df=pd.DataFrame(data);X=df[self.feature_columns+['position']].copy()

X['position_encoded']=self.position_encoder.transform(X['position']);X=X.drop('position',axis=1);y=df['rating']
        self.model.fit(X,y);joblib.dump(self.model,'xgboost_model.pkl');print(f"Model retrained with {len(data)} records")
```

Searching Algorithm:

The team code search algorithm implements a lookup mechanism to find a team object based on a unique identifier known as the team code. The algorithm accepts a team code input and searches through a predefined collection (such as an array or list) of team objects, each containing properties like teamCode and teamName.

A common approach is **linear search**, which iterates over each team in the collection, comparing the input code to the team's teamCode attribute. When a match is found, the corresponding team object is returned immediately, terminating the search. If no matching code is found after checking all entries, the algorithm returns a null or error state indicating the team does not exist.

For larger datasets, performance can be improved using a **binary search** on a sorted list of team codes or by employing a **hash map** (dictionary) to achieve average-case constant time lookups (O(1)).

This search method ensures efficient retrieval of teams without dependence on external databases or services, making it well suited for offline or client-side implementations.

```
function searchTeam() {
    const inputCode = document.getElementById("teamCodeInput").value.trim();
    const team = teams.find(t => t.code.toLowerCase() === inputCode.toLowerCase());
```

```
    const resultDiv = document.getElementById("result");
    if (team) {
        resultDiv.innerHTML = `<p>✅ Team Found: <b>${team.name}</b> (${team.members} members)</p>`;
    } else {
        resultDiv.innerHTML = `<p>❌ No team found with code "${inputCode}"</p>`;
    }
}
```

Chapter 4 : Implementation and Testing

4.1 Implementation

The implementation phase involved the practical development of the Athletix web application. It includes the setup of the development environment, selection of appropriate technologies, and integration of frontend, backend, and database components to deliver the required functionality in a web environment.

4.1.1 Tools Used

Frontend Tools

The front end of this Web application is developed using **HTML and CSS** for layout design and **JS** for implementing application logic. The development was carried out using **VScode** code editor.

- **HTML (HyperText Markup Language) and CSS (Cascading StyleSheet)** is used to define the user interface components of the web-app. Each screen is built using HTML files, ensuring structured design and proper control hierarchy.
- **Javascript** serves as the programming language for writing application logic. It handles user interaction, event handling, and communication between the app and backend services. Js is one of the most popular languages for Web development and is known for platform independence.
- **VScode** provides built-in tools like real-time preview, debugging, and layout editor, allowing efficient development and testing of the UI components.

Backend and Database

The backend of Athletix is implemented using **Firebase** for real-time cloud-based data storage and user authentication. For data access and caching, **Firebase**, a platform developed by Google, offers several essential features including:

- **Authentication**: Secure user login with email and password.

- **Realtime Database**: Cloud-hosted NoSQL database to store team, match, and player data with real-time synchronization across devices.

- Firebase is a scalable solution that handles user data and keeps all connected devices updated automatically, making it ideal for real-time sports team management apps.

This data handling approach ensures that Athletix is functional only online.

Documentation Tools

- **Microsoft Word** was used for writing and organizing project documentation. Its powerful formatting and referencing tools helped create a clean and professional-looking report.
- **Canva** was used to create diagrams such as Use-Case, Class, Sequence, and Activity Diagrams. Its drag-and-drop functionality allows quick and efficient creation of standardized software modeling diagrams.
- **Visual Studio Code (VS Code)** was used occasionally for editing scripts, managing backend logic, and reviewing JSON structure from Firebase exports. It provides an advanced coding environment with support for version control and syntax highlighting.

4.2 Testing

4.2.1 Test Cases for Unit Testing

Table 1 : Test Case for user registration

S.N	Input User Data	Expected Result	Actual Result	Remarks
1	bishals01magar@gmail.com Admin@123	Registered successfully	Registered successfully	Pass
2	dibooshnepal0@xyz.cobm abcd12	Show Error	Error Displayed	Fail
3	abc@gmail.com 1234	Show Error	Error Displayed	Fail

Table 2: Test Case for user login

S. N	Input User Data	Expected Result	Actual Result	Remarks
1	bishals01magar@gmail.com Admin@123	Open Dashboard	Dashboard displayed	Pass
2	dwash@xyz.com abcd12	Show Error	Error Displayed	Fail
3	abc@gmail.c om 1234	Show Error	Error Displayed	Fail

Table 3: Test Cases for System Testing

S. N	Input User Data	Expected Result	Actual Result	Remarks
1	Jhapa FC Jhapa Kathmandu UTC +5:45	Create Team	Team Created	Pass
2	10:45 AM VS IlamFC Domalal , Btm	Create Event	Event Created	Pass
3	3467282	Join Team	Team Joined	Pass

Chapter 5 : Conclusion and Future Recommendations

5.1 Conclusion

It has been a rich learning and satisfaction experience to be able to contribute towards the creation of this project titled AthletiX: A Football Team Management System. This web system was meant to serve football team managers and coaches by providing one system for scheduling, attendance tracking, team communication, and player performance monitoring.

The web application was programmed with HTML, CSS, and JavaScript for front-end programming to create a neat, responsive, and user-friendly interface from any web-connected device. Firebase provided cloud authentication, real-time database updating, and secure data storage so that communication could be smooth and instant synchronization of schedules, messages, and players' profiles was ensured.

To satisfy the goals of the project, we first thoroughly examined the functional needs and built a user-friendly interface to ensure easy navigation. Features of player registration and monitoring of profiles were implemented through Firebase Realtime Database, and scheduling and reminders were implemented by event-based scripts in JavaScript. The in-app messaging was made possible through Firebase's real-time messaging, where updates were transmitted instantly to all logged-in users.

In the process, we gained valuable experience in web application coding, database integration, and responsive UI design. Not just was this project a technical booster, but it also enhanced our problem-solving ability to create effective real-world software solutions that would satisfy end-users' expectations.

5.2 Lessons Learnt / Outcome

The primary goal of this project was to create a system that simplifies and automates football team management. The app allows coaches to manage teams, schedule games and practices, track player attendance, send real-time updates, and analyze performance through visual data representations.

Through this project, we learned to:

- Apply class based design in web development.
- Use Firebase for authentication and real-time data synchronization.

- Improve UI design using HTML, CSS and JS for better usability and user experience.
- Analyze project requirements and transform them into working features.
- Plan, structure, and document an application in a professional and academic manner.

The final system delivers core functionality as planned and meets the operational expectations of a coach or manager in both grassroots and competitive football environments. It significantly reduces manual communication and planning efforts, improving the efficiency of team coordination.

5.3 Future Recommendations

The success of Athletix will largely depend on how effectively coaches, players, and teams adopt and engage with the platform. While the current version of the app fulfills essential team management functions, there remains a vast scope for enhancement.

In the future, the system can be expanded with the following recommendations:

- **Role-Based Access Control**: Adding more user roles (e.g., Coach, Medical Staff) with specific permissions to improve collaboration.

- **Match Lineup and Formation Builder**: Interactive tools for planning player positions and tactical formations.

- **Player Profile Enhancements**: Include injury history, training logs, and personal achievements for long-term tracking.

- **Integration with Wearable Devices**: Allowing real-time performance tracking using GPS and fitness devices.

- **In-App Feedback & Analytics**: Track user activity and gather feedback to continuously improve the app's usability.

This project has laid a strong foundation for a practical and scalable football team management platform. With continued development, feedback from users, and support from potential stakeholders, Athletix can evolve into a comprehensive solution used widely by clubs at every level.

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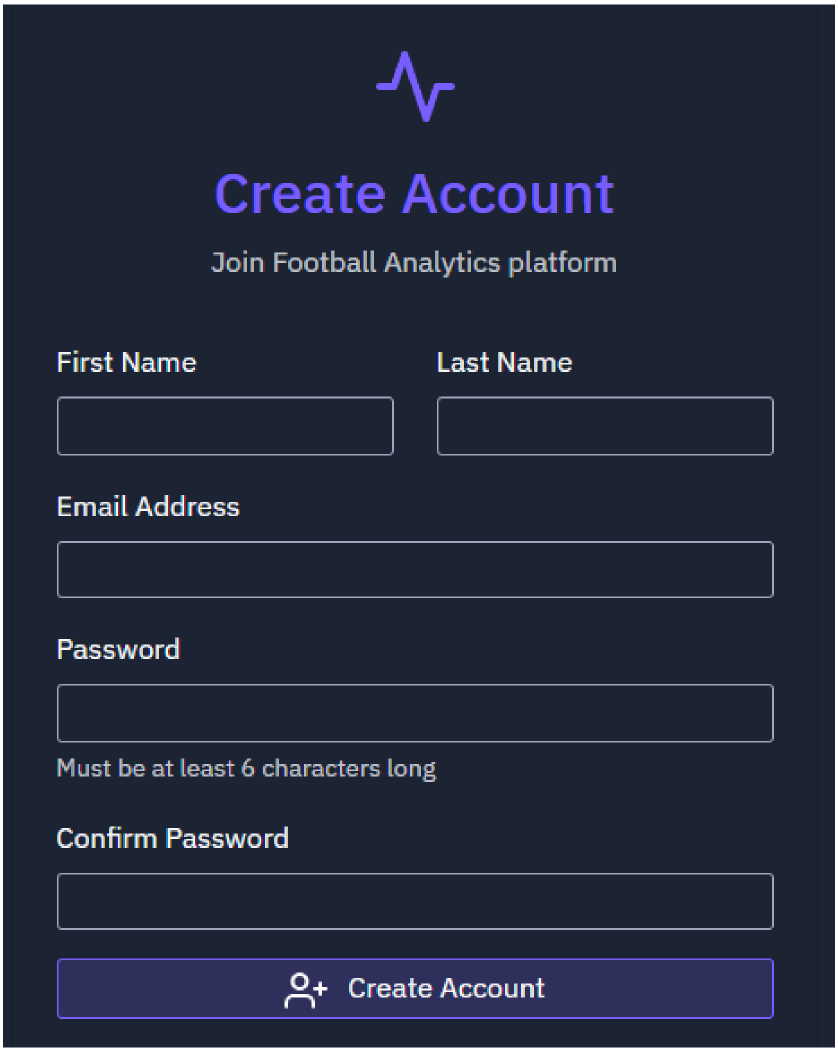
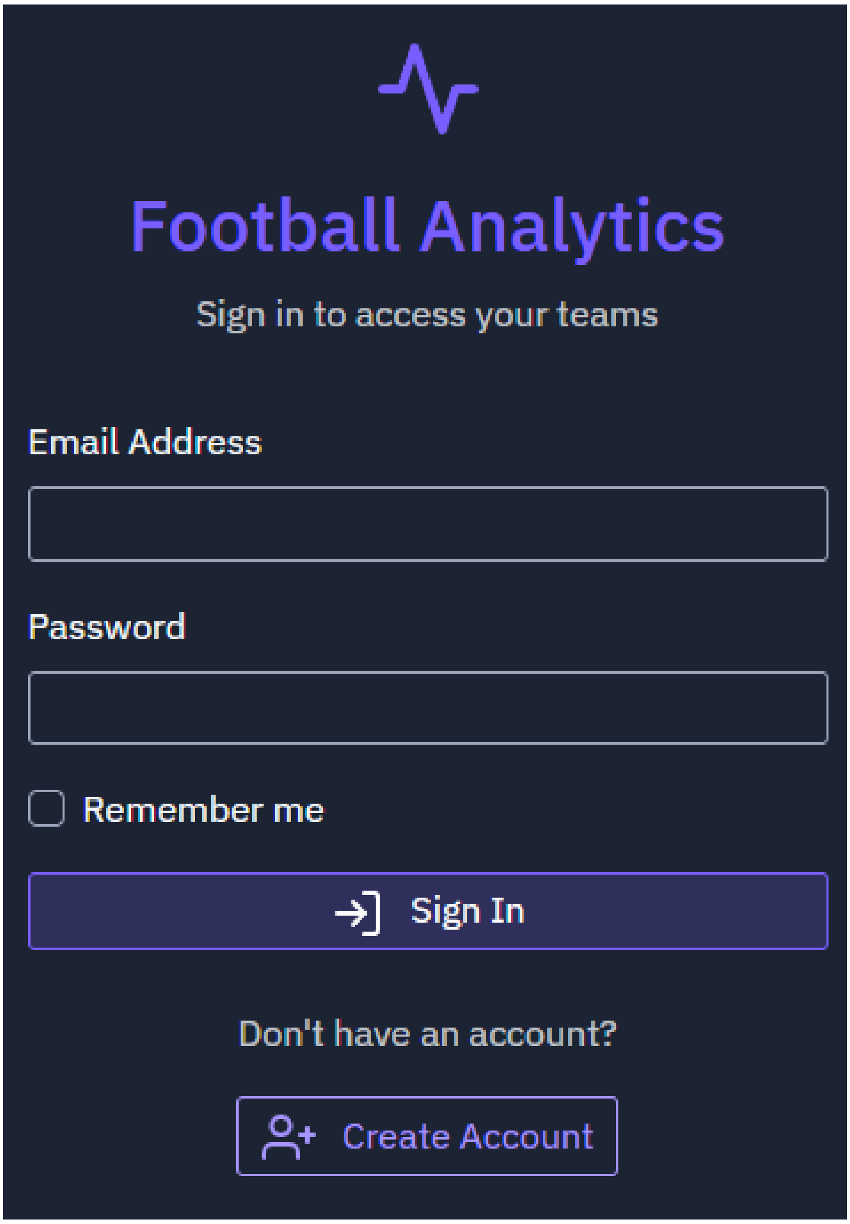
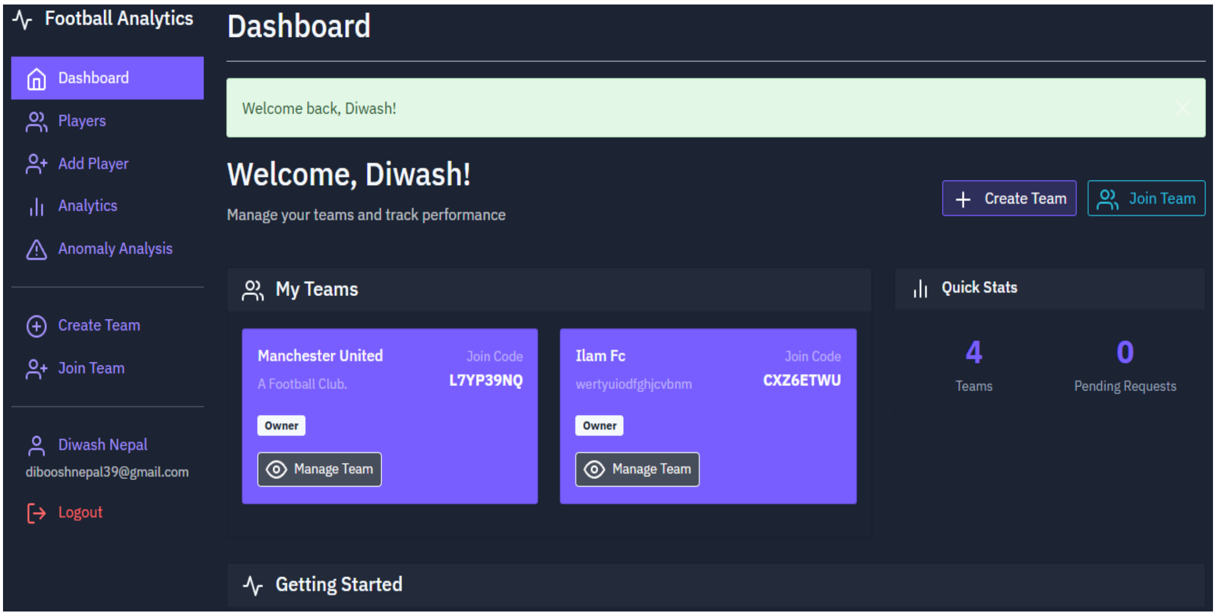
Appendices



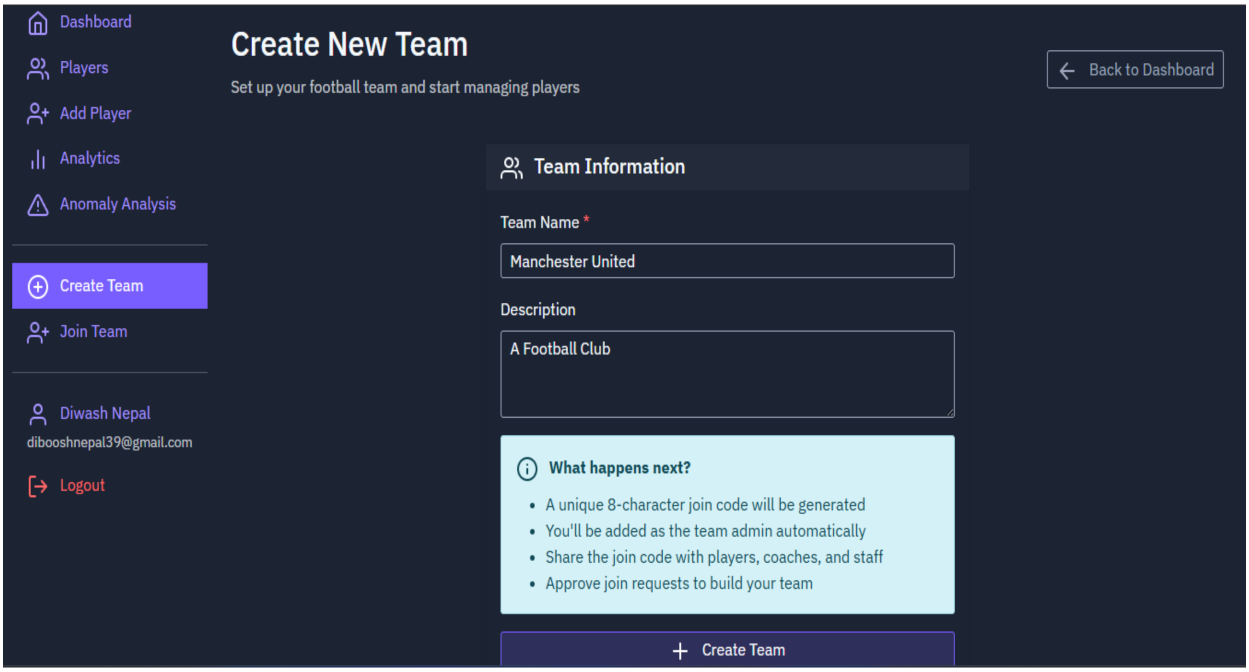
<u>Page-1: Signup</u>



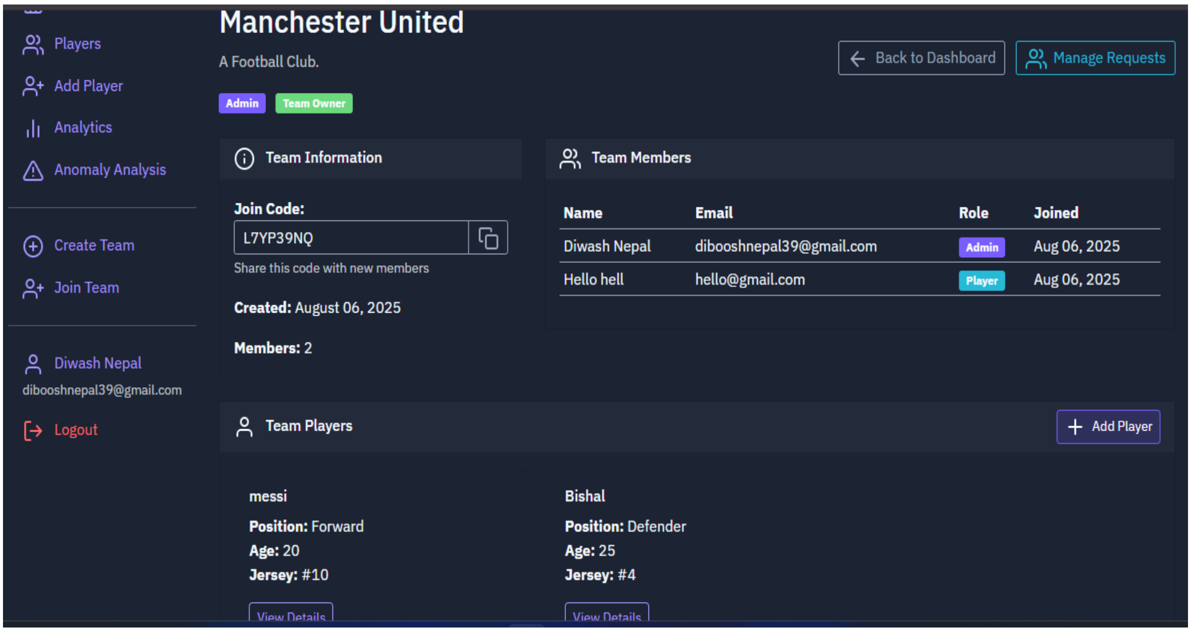
Page 2: SignIn/Login



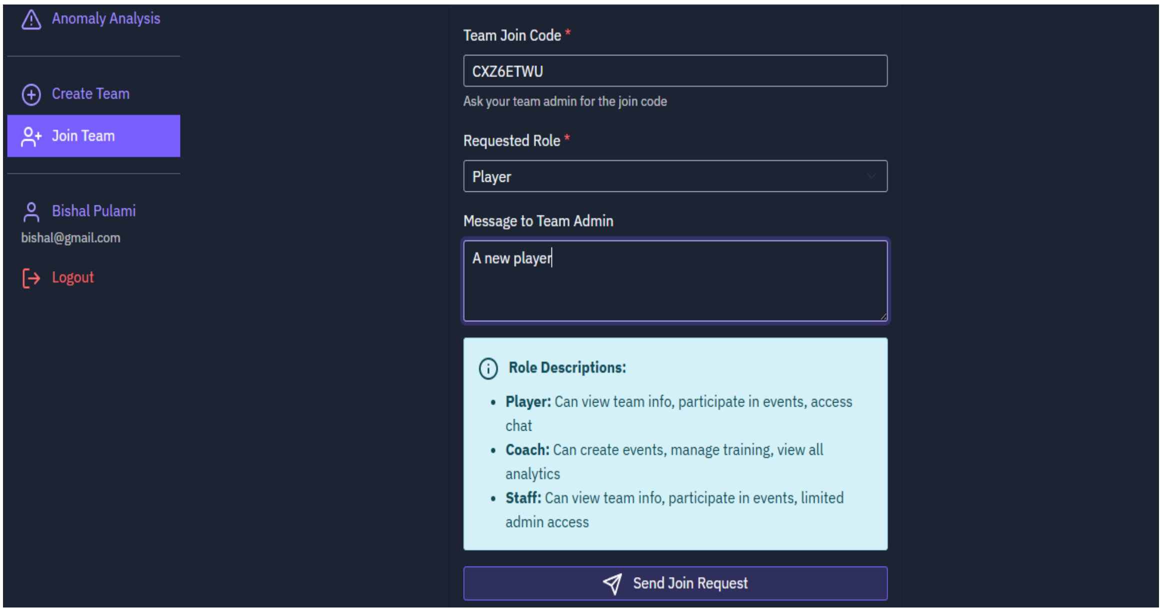
Page-3: Dashboard



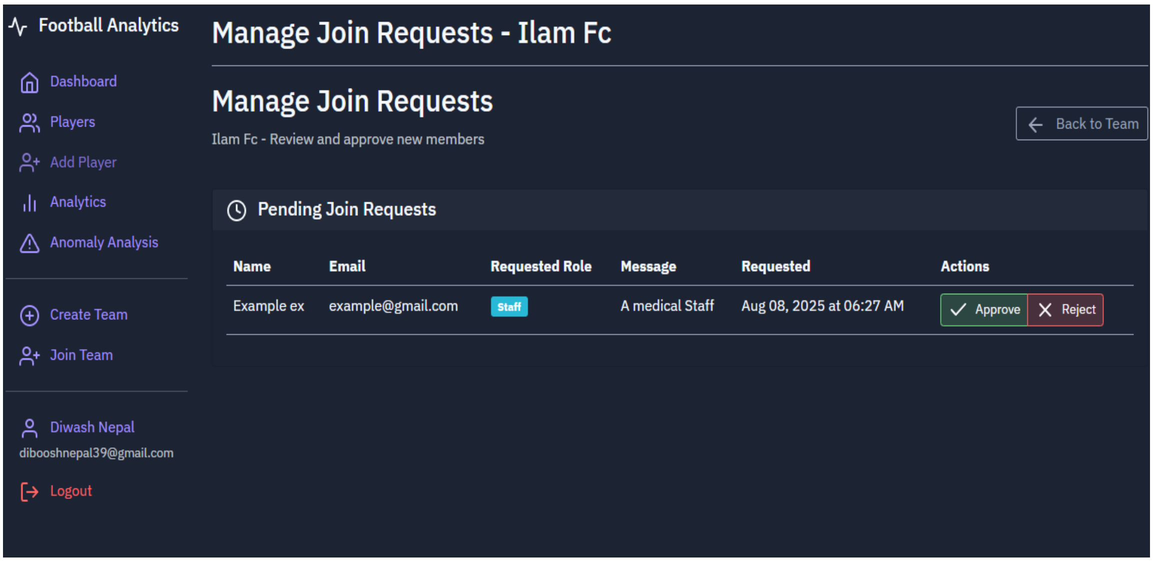
Page-4: Create Team



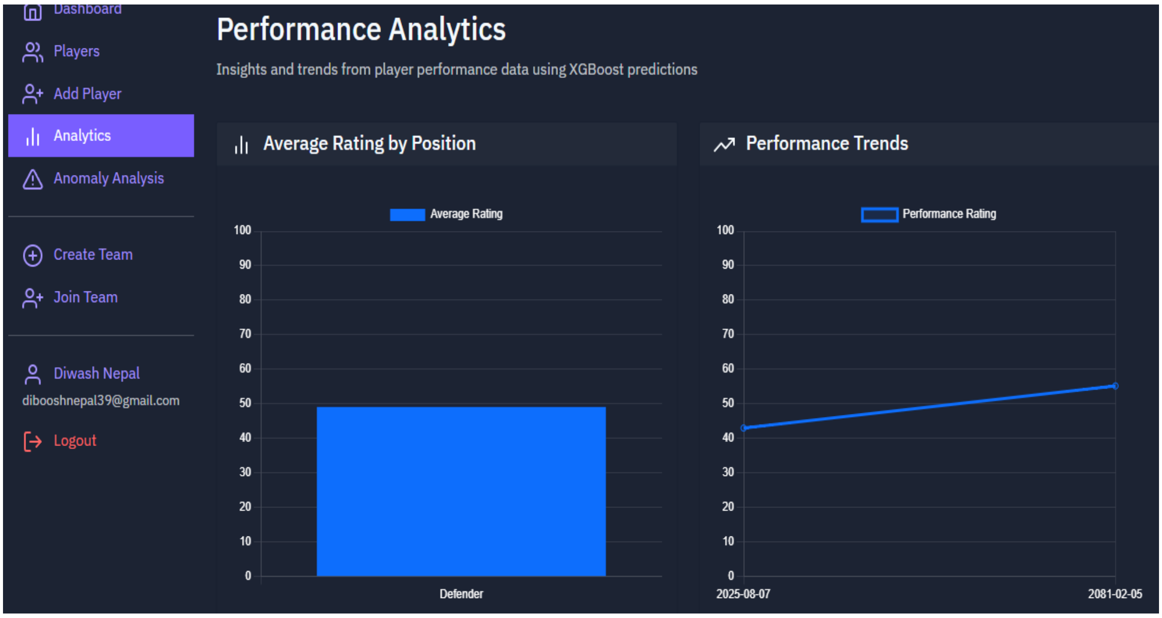
Page-4: Manage Team



Page-5: Join Team



Page-6: Join Request to admin



Page-7: Player Performance Analysis